

Portfolio3_Kabduali_Ayaulym

Synopsis: Correlation and Factor Analysis of Project Risk Factors

Objective:

The objective of this analysis is to identify and group key factors influencing project risks and performance using correlation and factor analysis techniques.

Data:

The dataset consists of multiple variables related to project risks, including operational issues, planning challenges, financial constraints, scope changes, external disruptions, and organizational characteristics.

Method:

Correlation analysis was conducted to examine relationships between variables. Based on the correlation matrix, variables were grouped into underlying factors using factor analysis. Each factor represents a cluster of related issues affecting project performance.

Results:

The analysis identified several key factors. The most significant groups include operational and on-site execution issues, design and planning challenges, and financial constraints. Additional factors include scope changes, external disruptions, coordination challenges, and organizational characteristics.

However, the overall factor structure appears relatively weak, indicating that some variables may not be strongly correlated or may overlap across multiple factors.

Conclusion:

Factor analysis helps simplify complex datasets by grouping related variables into meaningful categories. Despite some limitations in structure, the analysis provides useful insights into the main sources of project risks and can support better decision-making.